

Lemma.

Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F . Then, for any $\sigma \in \text{Aut}(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial for α over F .

Proof. Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in F[x]$ be the minimal polynomial for α . Then, $f(\alpha) = 0$, so $\sigma(f(\alpha)) = 0$. (The field homomorphism preserves zero).

Now, $0 = \sigma(f(\alpha)) = \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0)$

$$\begin{aligned} \sigma \text{ fixes } F \quad \hookrightarrow &= (\sigma\alpha)^n + \sigma(a_{n-1}) \cdot (\sigma\alpha)^{n-1} + \dots + \sigma(a_1) \cdot (\sigma\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1(\sigma\alpha) + a_0 \\ &= f(\sigma\alpha). \end{aligned}$$

Prop. Let $H \leq \text{Aut}(K)$ be a subgroup of the group of all automorphisms of K .

Then, the collection F of elements of K fixed by all the elements of H is a subfield.

Proof. First, given $\sigma \in \text{Aut}(K)$, the set $F_\sigma = \{a \in K \mid \sigma(a) = a\}$ is subfield, because:

Let $a, b \in F_\sigma$ be given. Then, $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$, and if $b \neq 0$, then $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$, so the claim is proved.

And now, since the arbitrary intersection of fields is field, so proved. ~~QED~~
Such a subfield is called the fixed field of $S \subseteq \text{Aut}(K)$.

Prop. Let K be a field. Then,

If $F_1 \leq F_2 \leq K$, then $\text{Aut}(K/F_2) \leq \text{Aut}(K/F_1)$.

If $H_1 \leq H_2 \leq \text{Aut}(K)$, then $F_{H_2} \leq F_{H_1}$.

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Prop. Let E be the splitting field over F of the polynomial $f(x) \in F[x]$.
Then, $|\text{Aut}(E/F)| \leq [E:F]$, and the equality holds if $f(x)$ is separable over F .
Proof. Clearly, E/F is finite extension.